02170 Database Systems - Written TEST Exam Part 2, 2026

Der anvendes en scoringsalgoritme, som er baseret på "One best answer"

Dette betyder følgende:
Der er altid netop ét svar som er mere rigtigt end de andre
Studerende kan kun vælge ét svar per spørgsmål
Hvert rigtigt svar giver 1 point
Hvert forkert svar giver 0 point (der benyttes IKKE negative point)

The following approach to scoring responses is implemented and is based on "One best answer"
There is always only one correct answer – a response that is more correct than the rest
Students are only able to select one answer per question
Every correct answer corresponds to 1 point
Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points)

1. Conceptual Design

The conceptual design for the database *CompanyDatabase* was specified by an Entity-Relationship (E-R) diagram before the relational design (with the relational schemas shown on page 2 in Part 1 of this exam) was made.

We are now going to consider a part of this diagram. In order to answer the questions below, you might need to study the text and the tables on page 2 of Part 1 of this exam. For convenience, relevant pieces of that text and tables are repeated here:

Department(<u>dname</u>, country)

dname	country
Administration	Denmark
FormalSys	Denmark
Nobody	Utopia
SmartDB	Italy
SmartSys	Germany

Employee(<u>eid</u>, ename, dname)

eid	ename	dname
10	Anne	FormalSys
20	Giovanni	SmartDB
30	Ida	FormalSys
40	Jan	SmartSys
50	Signe	Administration

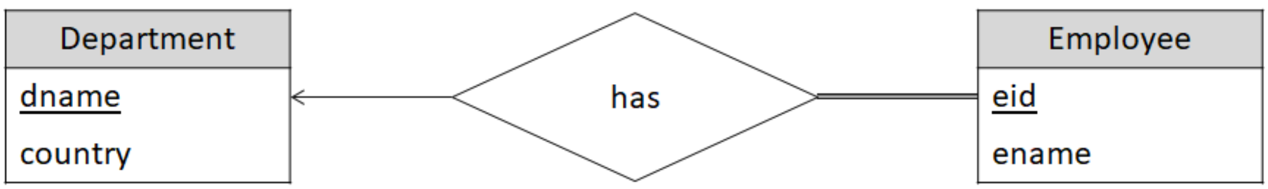
Question 1

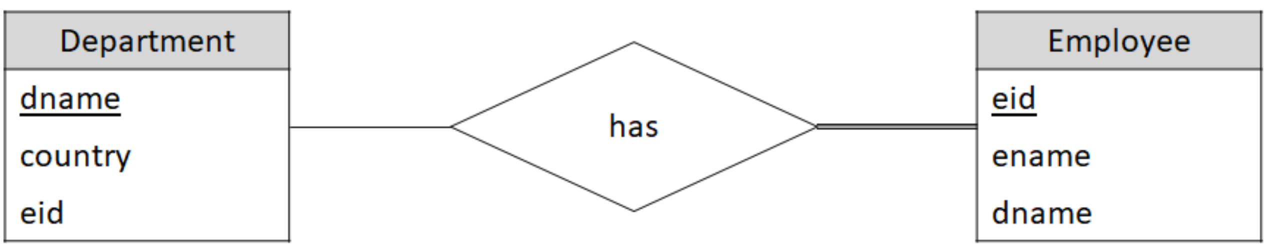
Which of the following (sub-)diagrams is a correct E-R diagram specifying departments, employees and the *has* relation between employees and departments?

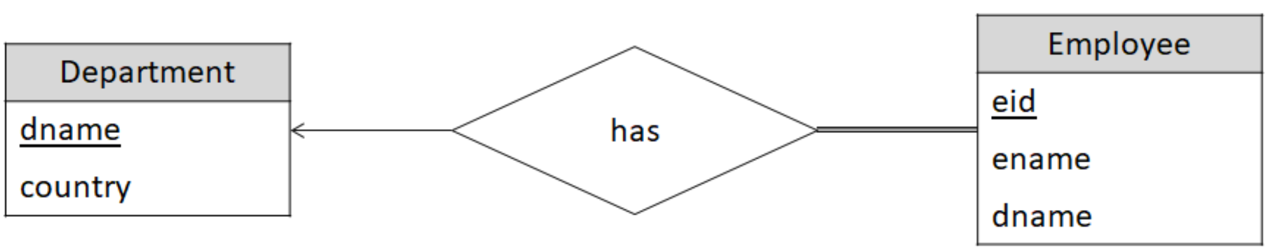
Choose one answer

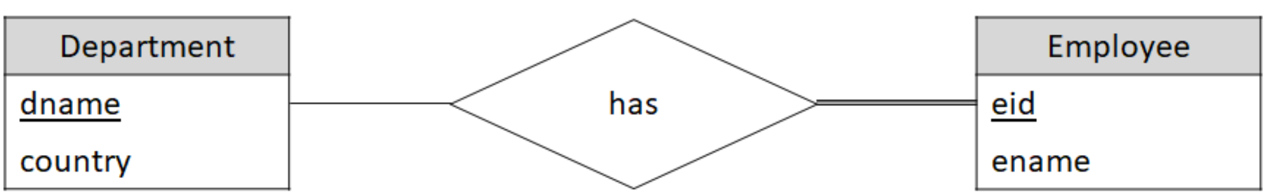
- ○ 

- ○ 

- ○ 

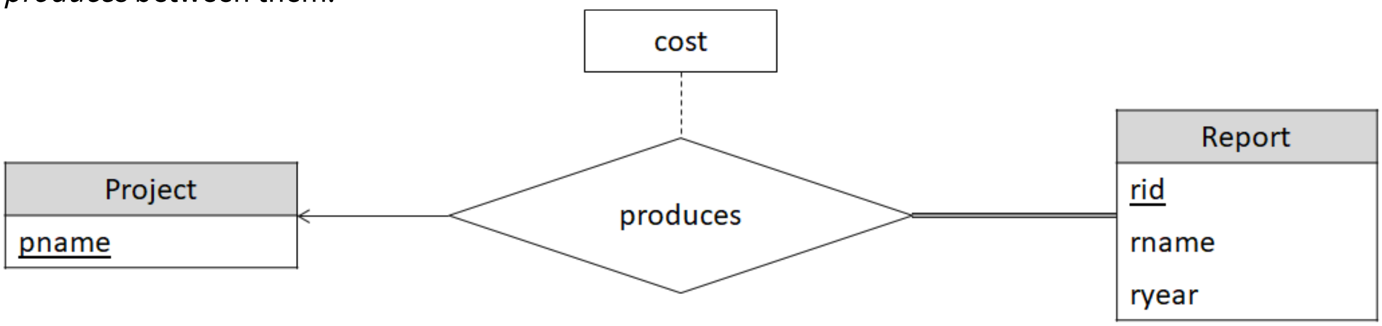
- ○ 

- ○ 

- ○ 

2. Entity-Relationship Diagrams and their Conversion to Relation Schemas

The database engineer is going to develop a database for a company to keep track of which reports were produced for which projects, and the cost incurred to produce them. The Entity-Relationship Diagram below is a conceptual model describing entity sets *Project* and *Report* and a relationship set *produces* between them:



Questions 2.1-2.4

The questions below are concerned with the implications of the depicted cardinality of the *produces* relation and the participation of *Project* and *Report* in *produces.*

Select the correct answers

	yes	no
Can two projects produce the same report?	○	○
Can a project produce no reports?	○	○
Can all reports produced for the same project have all the same cost?	○	○
Can a report be produced for no project?	○	○

Question 2.5

Convert the Entity-Relationship diagram shown above into relation schemas using the method described in the textbook and slides adopted in this course. What will the result be (here shown without foreign key constraints)?

Choose one answer

- ○ Project(pname, rid), Report(rid, rname, ryear, cost)
- ○ Project(pname, cost), Report(rid, rname, ryear), produces(pname, rid)
- ○ Project(pname, rid, cost), Report(rid, rname, ryear)
- ○ Project(pname), Report(rid, rname, ryear, pname, cost)

3. Normalization

Consider a relation with schema *R(A, B, C, D, E)*, where the primary key is not shown.

The following functional dependencies are assumed:

- $A, B \rightarrow C$
- $A \rightarrow D$
- $C \rightarrow E$

Question 3.1

Which of the following attribute sets are candidate keys for *R*?

Choose one answer

- ○ There are three candidate keys: {A, B} , {A} and {C}
- ○ There is one candidate key: {A, B, C}
- ○ There is one candidate key: {A, B}

Question 3.2

What is the highest normal form of the relation R with the schema shown above?

Choose one answer

- 2NF
- 1NF
- 3NF

Question 3.3

Choose a primary key for the schema *R* and perform normalization of *R* to at least 3NF (using the method described in the textbook adopted in this course). What is the result of that (ignoring foreign key constraints)?

Choose one answer

- R1($\underline{B}$, D), R2($\underline{C}$, E), R3($\underline{A}$, $\underline{B}$)
- R1($\underline{C}$, E), R2($\underline{A}$, $\underline{B}$, C, D)
- R($\underline{A}$, $\underline{B}$, C, D, E)
- R1($\underline{A}$, D), R2($\underline{C}$, E), R3($\underline{A}$, $\underline{B}$, C)
- R($\underline{A}$, $\underline{B}$, $\underline{C}$, D, E)
- R1($\underline{A}$, D), R2($\underline{A}$, $\underline{B}$, C, E)

Questions 3.4-3.5

Consider a relation with schema *S(A, B, C)* for which the following multi-valued functional dependencies are assumed:

- A →→ B
- A →→ C

Decide for each of the following relation instances (tables) whether it satisfies the stated dependencies.

Select the correct answers

Table	yes	no
A B C / a1 b1 c1 / a1 b1 c2 / a1 b2 c1 / a1 b2 c2	○	○
A B C / a1 b1 c1 / a1 b1 c2 / a1 b2 c1 / a1 b3 c3	○	○

A	B	C
a1	b1	c1
a1	b1	c2
a1	b2	c1
a1	b2	c2

A	B	C
a1	b1	c1
a1	b1	c2
a1	b2	c1
a1	b3	c3

4. Integrity Constraints

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Questions 4.1-4.3

Departments (identified by their unique name *dname*) may have local sports clubs (identified by a local club id *cid*). Consider the following schema for a relation containing information about the price for a membership of each club of each department.

Club(<u>dname</u>, <u>cid</u>, price)

The following questions concern which relation/table instances are legal for this logical design.

Select the correct answers

	yes	no
Can two rows have the same *dname* and the same *price*, but different *cid*?	○	○
Can two rows have the same *dname* and the same *cid*, but different *price*?	○	○
Can two rows with different *dname* have the same *cid?*	○	○

Consider the following relation schemas:

- RA(A, B, E) **FOREIGN KEY** (B) **REFERENCES** RB(B), **FOREIGN KEY** (E) **REFERENCES** RE(E)
- RB(B, C) **FOREIGN KEY** (C) **REFERENCES** RC(C)
- RC(C, D)
- RE(E, F)

Assume that these have been implemented in SQL and all three foreign key constraints have been given the referential action **ON DELETE CASCADE.**
Which tables would *potentially* be changed when a row is deleted in the *RC* table*?*
(Potentially means that it will happen for some table instances, but not for others.)

Select the correct answers

	yes	no
Is *RA* potentially changed?	○	○
Is *RB* potentially changed?	○	○
Is *RE* potentially changed?	○	○

5. Indexing and Hashing

When answering the questions on this page, you should use the terminology and theory of the text book and slides adopted in this course.

In these questions it is assumed that the database system does not automatically choose any indexes.

In the first two questions, you will be asked about the type of the following index for the ***Department***(*<u>dname</u>*, *country*) relation with search key *dname*:

dname
Administration
SmartDB

dname	country
Administration	Denmark
FormalSys	Denmark
Nobody	Utopia
SmartDB	Italy
SmartSys	Germany

Question 5.1
Is it primary or secondary?

Choose one answer

- secondary
- primary

Question 5.2

Is it dense or sparse?

Choose one answer

- ○ sparse
- ○ dense

Question 5.3

Assume that the database management system does not automatically choose indexes, and that a database statistics shows that a very common query is of the form

SELECT rid **FROM** Report **WHERE** rname = x,

where *x* is some report name.

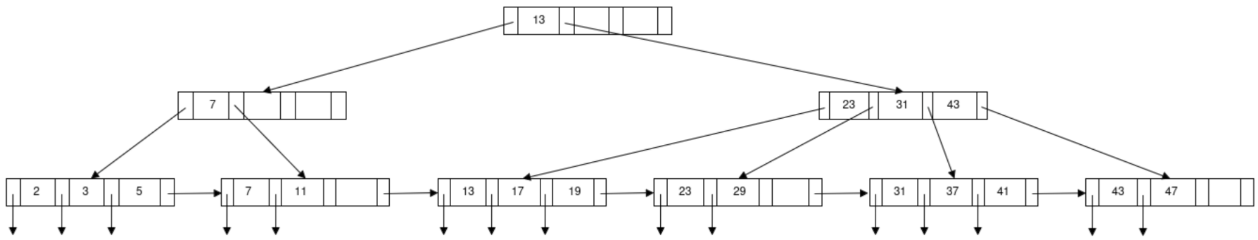
Under these assumptions, which of the following would you do to make these queries faster?

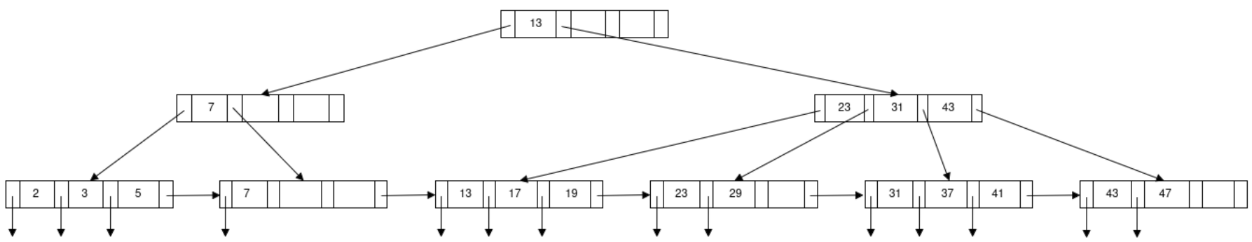
Choose one answer

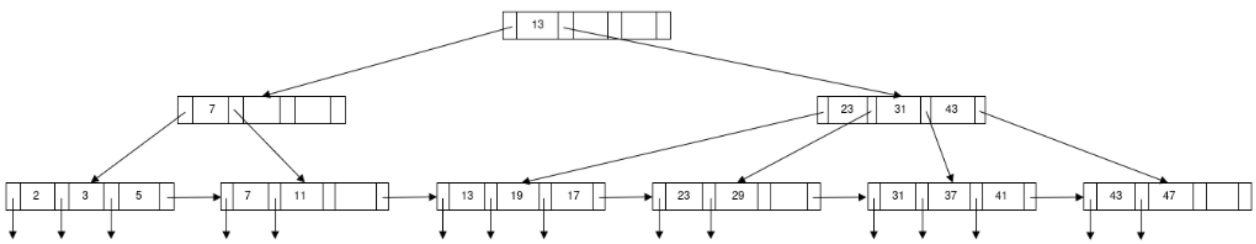
- ○ I would make an index on *rname*
- ○ I would use hashing with search key *rid*
- ○ I would use hashing with search key *rname*.
- ○ I would make an index on *rid*

Question 5.4
Which of the following trees is a legal B+ tree?

Choose one answer

- ◯ 

- ◯ 

- ◯ 

6. Formal Relational Query Languages

The following questions consider the following relation schemas of *CompanyDatabase* described in Part 1 of this exam:

Employee(eid, ename, dname),
authors(rid, eid)
participates(pname, eid*)*

Question 6.1

Relational Algebra: Which of the following is a correct Relational Algebra expression finding the *eid* of those employees who are authors of at least one report, but do not participate in any project?

Choose one answer

- ◯ $\Pi_{eid}(authors) - \Pi_{eid}(participates)$
- ◯ $\Pi_{eid}(authors) \cap \Pi_{eid}(participates)$
- ◯ $\Pi_{eid}(authors - participates)$
- ◯ $\Pi_{eid}(authors) \wedge \neg\Pi_{eid}(participates)$

Question 6.2

Domain Calculus: Which of the following is a correct Domain Calculus expression finding for each employee who never authored any report, his/her *ename* and *dname*?

Choose one answer

- $\{\langle (en, dn) \rangle | \langle e, en, dn \rangle \in Employee \wedge \neg(\exists r(\langle r, e \rangle \in authors))\}$
- $\{\langle (en, dn) \rangle | \exists e, en, dn(\langle e, en, dn \rangle \in Employee \wedge \neg(\exists r(\langle r, e \rangle \in authors)))\}$
- $\{\langle (en, dn) \rangle | \exists e(\langle e, en, dn \rangle \in Employee \wedge \neg(\exists r(\langle r, e \rangle \in authors)))\}$
- $\{\langle (en, dn) \rangle | \exists e(\langle e, en, dn \rangle \in Employee) \wedge \neg(\exists r(\langle r, e \rangle \in authors))\}$